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标题	Pyrene Functionalized Highly Reduced Graphene Oxide-palladium Nanocomposite: A Novel Catalyst for the Mizoroki-Heck Reaction in Water
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摘要（原文节选）

The formation of a C-C bond through Mizoroki-Heck cross-coupling reactions in water with efficient heterogeneous catalysts is a challenging task. In this current study, a highly reduced graphene oxide (HRG) immobilized palladium (Pd) nanoparticle based catalyst (HRG-Py-Pd) is used to catalyze Mizoroki-Heck cross-coupling reactions in water. During the preparation of the catalyst, amino pyrene is used as a smart functionalizing ligand, which offered chemically specific binding sites for the effective and homogeneous …

数据表 · TABLE 1（15 行，含数值行 15 行）

Mizoroki-Heck reaction (MH), 4-bromoanisol and acrylic acid as a model substrate. Catalyst screening.

Sl. No	Catalyst	Cat. Wt. (mg)	K3PO4 (eq.)	SDS (Eq.)	Temp (oC)	HPLC Conversion
1	Pd-NPs	5.3	2	0.5	R.T.	—
2	5.3	2	0.5	90 ° C	traces	
3	3.0	2	0.5	100 ° C	40%	
4	5.3	2	0.5	100 ° C	92%	
5	8.0	2	0.5	100 ° C	74%	
6	HRG-Pd	5.3	2	0.5	R.T.	—
7	5.3	2	0.5	90 ° C	—	
8	3.0	2	0.5	100 ° C	—	
9	5.3	2	0.5	100 ° C	—	
10	8.0	2	0.5	100 ° C	—	
11	HRG-Py-Pd	5.3	2	0.5	R.T.	—
12	5.3	2	0.5	90 ° C	traces	
13	3.0	2	0.5	100 ° C	93%	
14	5.3	2	0.5	100 ° C	84%	
15	8.0	2	0.5	100 ° C	71%	

脚注（逐字）：Reaction condition: 4-bromoanisol (0.5 mmol) and acrylic acid (0.5 mmol), catalyst (3.0 – 8.0 mg), K3PO4 (1.0 mmol), SDS (0.25 mmol) and water (2 ml) at R.T.—100 ° C, time 20 h.

正文对应句（逐字）：After examining the efficiency of all the as-prepared nanocatalysts at different catalyst loading, the effect of reagents (SDS, K3PO4), solvent, and reaction time were evaluated on the same model MH reaction, using the best performing catalyst mass loading condition i.e. Pd-NPs (5.3 mg) and HRG-Py-P

数据表 · TABLE 2（26 行，含数值行 22 行）

Mizoroki-Heck reaction of 4-bromoanisol (1a) and acrylic acid (2) in aqueous solution using best catalyst HRG-Py-Pd; Optimization for K3PO4, SDS, and reaction time.

Sl. No	Catalyst	Cat. Wt. (mg)	K3PO4 (Eq.)	SDS (Eq.)	Time (h)	HPLC Conversion
SDS optimization						
1	Pd-NPs	5.3	2	—	20	—
2	5.3	2	0.25	20	19%	
3	5.3	2	0.5	20	92%	
4	5.3	2	1.0	20	88%	
5	HRG-Py-Pd	3.0	2	—	20	—
6	3.0	2	0.25	20	24%	
7	3.0	2	0.5	20	93%	
8	3.0	2	1.0	20	92%	
K3PO4 Optimization						
9	Pd-NPs	5.3	0.75	0.5	20	20%
10	5.3	1.0	0.5	20	37%	
11	5.3	1.5	0.5	20	78%	
12	HRG-Py-Pd	3.0	0.75	0.5	20	68%

Sl. No	Catalyst	Cat. Wt. (mg)	K3PO4 (Eq.)	SDS (Eq.)	Time (h)	HPLC Conversion
13	3.0	1.0	0.5	20	98%	
14	3.0	1.5	0.5	20	97%	
Reaction Time Optimization						
15	Pd-NPs	5.3	2.0	0.5	2	20%
16	5.3	2.0	0.5	3	36%	
17	5.3	2.0	0.5	8	62%	
18	HRG-Py-Pd	3.0	1.0	0.5	2	90%
19	3.0	1.0	0.5	3	99%	
20	3.0	1.0	0.5	8	98%	
Different Solvent System (DMF)						
21	Pd-NPs	5.3	2.0	0.5	20	-
22	HRG-Py-Pd	3.0	1.0	0.5	20	-

脚注（逐字）：Reaction condition: 4-bromoanisol 1a (0.5 mmol) and acrylic acid 2 (0.5 mmol), catalyst used Pd-NPs, 5.3 mg and HRG-Py-Pd 3.0, water (2 ml) at 100 ° C, time 2 – 20 h.

正文对应句（逐字）：After examining the efficiency of all the as-prepared nanocatalysts at different catalyst loading, the effect of reagents (SDS, K3PO4), solvent, and reaction time were evaluated on the same model MH reaction, using the best performing catalyst mass loading condition i.e. Pd-NPs (5.3 mg) and HRG-Py-P

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